

Week 01 — LangGraph Agent: *run_sql* + *think*

DSi AI Club · Individual assignment · Submit as a single .ipynb

Goal

Build a minimal LangGraph ReAct-style agent that can answer questions about a small database. The agent must expose two tools: **run_sql** for querying a database of your choice, and **think**, a hacky tool that captures the LLM's internal reasoning so even non-reasoning models can 'think out loud'.

The agent must use **MemorySaver()** as its checkpointer so that a single thread keeps short-term memory across turns. Conversations should **not** be persisted to a database — in-process memory only.

Required tools

- run_sql(query: str) -> str** — executes a read-only SQL statement against a database you set up locally. SQLite, DuckDB, Postgres, MySQL, or anything else is fine; pick whatever you're comfortable with. Return the rows as text the LLM can read.
- think(thought: str) -> str** — the 'hacky reasoning' tool. The LLM passes its current chain-of-thought as *thought*; the tool simply echoes back a confirmation string like *'Thought captured.'* Nothing more. The point is to give non-reasoning models an explicit slot to externalise reasoning steps before acting.

Constraints

- Use **LangGraph** (not bare LangChain agents).
- Checkpointer must be **MemorySaver()**. Do not wire SqliteSaver or any DB-backed checkpointer.
- Pick any LLM provider you have access to (OpenAI / Anthropic / Groq / Ollama...).
- Seed the database with a few rows so the agent has something interesting to query.
- Demonstrate at least one multi-turn interaction in the same thread to prove memory works.
- Show at least one trace where the agent calls *think* before *run_sql*.

Suggested deliverable shape

```
submissions/<empId>-<name>/week-01/  
  week-01.ipynb      # everything lives here  
  schema.sql         # optional, if you prefer a separate seed script
```

Keep everything in the notebook this week — agent code, DB setup, runs, and your narration. No *README*, no separate *.py* imports. The *library/* folder stays empty for week 01; graduation of reusable helpers into *library/* starts from later weeks, when a future notebook actually wants to reuse something from a previous one.

What we'll look at

- Does the agent actually call both tools, in sensible order?
- Does *think* capture real intermediate reasoning, not just a restatement of the question?
- Does memory survive across turns within one thread, and reset across threads?
- Is the notebook readable end-to-end, with at least one annotated example run?

Submission

Work on a feature branch — no direct pushes to *main*. Suggested branch name:
<empld>-<name>/week-01 (e.g. *539-apu/week-01*).

- Branch off *main*, commit your work under *submissions/<empld>-<name>/week-01/*.
- Open a Pull Request into *main* before the weekly deadline.
- Only edit files inside your own member folder — don't touch anyone else's.
- Keep secrets out of the notebook; use a local *.env* (already gitignored).